

src/app/impactHeatmap.ts

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Overview

`src/app/impactHeatmap.ts` adds a heat-map generator for district impact data.

- Imports `RiskSignal` from `./types` (line 1).
- Declares ``District`

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`Score` (lines 3-10) with `district`, `score`, `changedPaths`, `riskCount`, and `highSeverity``.

- Provides ``buildDistrict`

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`Heatmap`` (lines 17-67) to aggregate scores from changed paths, risk signals, and affected node IDs.

- Exposes `heatmapMermaid` (lines 68-79) and `heatmapHtmlTable` (lines 80-93) for visual output.
- Includes `escapeHtml` (lines 94-100) for safe HTML rendering.

Key changes

- **Interface*
- – ``District`

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`Score` now tracks `highSeverity` and `riskCount`` (lines 3-10).

- **Scoring logic*
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- Each changed path adds 2 points and increments `changedPaths` (line 40).
- Risk signals are weighted: high = 8, medium = 4, low = 2; they increment `riskCount` and `highSeverity` (lines 46-50).
- Affected node IDs add 1 point per node (lines 55-61).
- Results are sorted by descending `score` (line 65).

- **Visualization*
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- `heatmapMermaid` builds a Mermaid flowchart, limiting to `maxRows` districts (lines 68-79).
- `heatmapHtmlTable` renders an HTML table with a proportional impact bar (lines 80-93).
- **Safety*
- – `escapeHtml` sanitizes district names for HTML output (lines 94-100).

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- **Granularity*
- – Explicit risk-severity weighting (lines 46-50) provides finer impact distinctions.
- **Centralization*
- – Aggregation logic resides in a single function (``buildDistrict`

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Heatmap`), reducing duplication.

- **Observability*
- – Mermaid and HTML outputs enable quick visual checks in dashboards or docs.
- **Performance*
- – Operations are linear or $n \log n$ over the number of districts, suitable for typical data sizes.

Risks & follow-ups

- `highSeverity` is recorded but not displayed; confirm if downstream consumers need this field.
- Node-ID parsing assumes `file:` prefixes or colon-separated paths (lines 55-58); test against all node-ID formats.
- Existing components that consume district impact data must import the new interface and functions.
- Add unit tests for edge cases (e.g., no changed paths, all high-severity risks) to guard against regressions.